PHYSIOLOGY

Water & Electrolyte Balance by the Kidneys

LG 1
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GUE BLOCK

Physiology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

1. What are the daily intake and output routes for water, and roughly how much does each contribute?
2. Walk through the 5-step ADH cascade from the stimulus to the urine result.
3. Which nephron segment is impermeable to water, and what does it reabsorb instead?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 The principle — intake must equal output

Water balance	Daily amount
Intake: drinks	≈1,200 mL/day
Intake: food	≈1,000 mL/day
Intake: metabolic water	≈300 mL/day
Total intake	**≈2,500 mL/day**
Output: urine	≈1,500 mL/day
Output: skin (sweat)	≈500 mL/day
Output: lungs	≈400 mL/day
Output: feces	≈100 mL/day
Total output	**≈2,500 mL/day**

KEY EQUATION

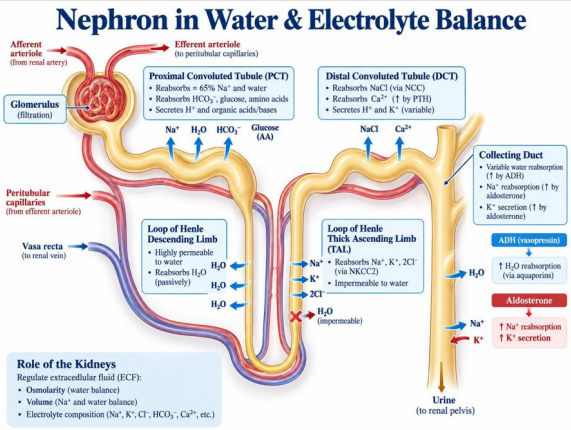
Renal excretion = Filtration − Reabsorption + Secretion. Example: during dehydration, a healthy kidney reduces urine volume and increases urine concentration.

2 Body fluid compartments & tonicity

Term	Definition
Total body water (TBW)	All body water — ≈60% of body mass in a typical young adult male
Intracellular fluid (ICF)	Fluid inside cells — ≈⅔ of TBW
Extracellular fluid (ECF)	Fluid outside cells — ≈⅓ of TBW
Osmolality	Number of dissolved particles per kg of water (mOsm/kg)
Tonicity	Effective osmotic force that changes cell volume

Major ECF cation: **Na^+**. Major ICF cation: **K^+**.

3 Nephron segment-by-segment handling



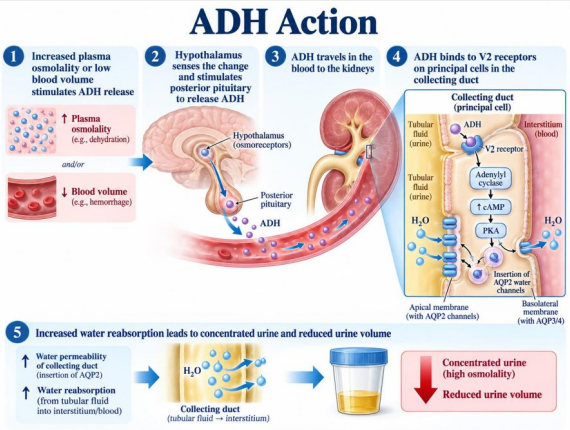
Segment	Handles
Proximal tubule (PCT)	≈65% of Na^+ and water; reabsorbs HCO_3^-, glucose, amino acids; secretes H^+ and organic acids/bases
Loop of Henle — descending limb	Highly permeable to water; reabsorbs H_2O passively
Loop of Henle — thick ascending limb (TAL)	Reabsorbs Na^+, K^+, $2Cl^-$ via NKCC2; **impermeable to water**
Distal tubule (DCT)	Reabsorbs NaCl (via NCC) and Ca^{2+} (↑ by PTH); secretes H^+ and K^+ (variable)
Collecting duct	Variable water reabsorption (↑ by ADH); Na^+ reabsorption and K^+ secretion (both ↑ by aldosterone)

Reabsorption: tubule → interstitium → peritubular blood (e.g. Na^+ and water return to circulation). **Secretion**: peritubular blood → tubule (e.g. K^+ added to final urine). Water moves by osmosis; active Na^+ transport creates most of the driving gradients.

4 Countercurrent multiplier — the medullary gradient

Cortex ≈**300 mOsm/kg** → inner medulla up to ≈**1,200 mOsm/kg**. Descending limb is permeable to H_2O; ascending limb reabsorbs NaCl and is impermeable to H_2O. The **vasa recta** preserve the gradient; urea recycling contributes especially to the inner medulla.

5 ADH — final control of water excretion



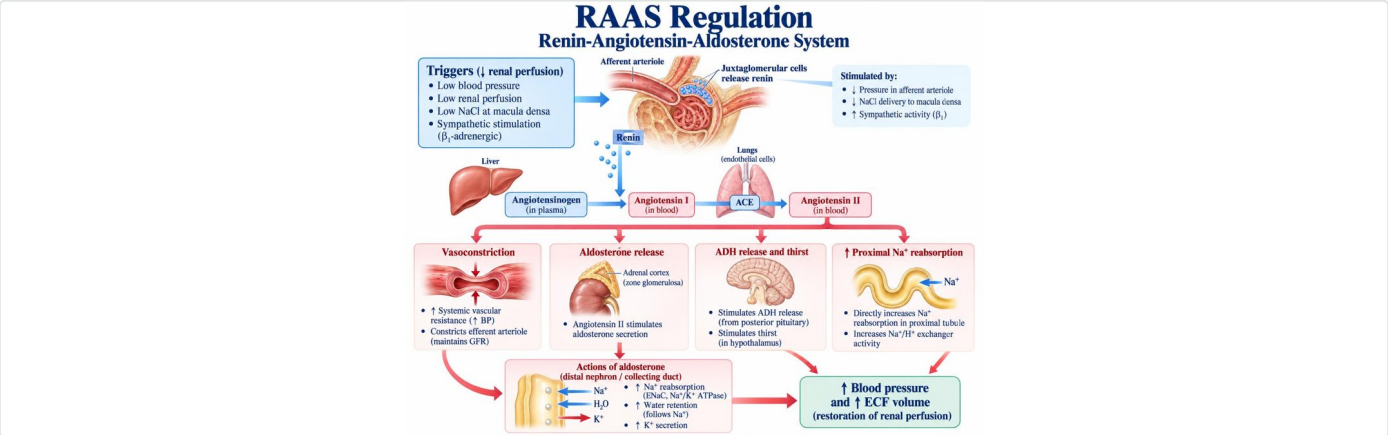
1	↑ Plasma osmolality or ↓ effective circulating volume
2	Hypothalamic osmoreceptors sense the change → posterior pituitary releases ADH
3	ADH binds the **V_2 receptor** on collecting-duct principal cells
4	**Aquaporin-2** channels are inserted into the luminal membrane
5	↑ Water reabsorption → ↓ urine volume, ↑ urine osmolality

ADH primarily regulates water; it does **not** directly "pump sodium" across the collecting duct.

State	ADH	Urine result
After large water intake	↓ ADH → fewer AQP2 channels	Large volume of **dilute** urine
Water deprivation	↑ ADH → more AQP2 channels	Small volume of **concentrated** urine

6 Sodium balance & RAAS

Fact	Detail
Sodium intake/ output	≈100–200 mEq/day (≈2–5 g/day); in steady state, intake = output and ECF volume/BP stay stable
Role of Na^+	Predominant ECF cation; total-body Na^+ mainly sets **ECF volume**, while plasma [Na^+] mainly reflects **water** balance relative to solute



1	↓ Renal perfusion / ↓ NaCl delivery to macula densa → triggers **renin** release from juxtaglomerular cells
2	Renin → angiotensin I → (ACE, lungs) → **angiotensin II**
3	Ang II constricts vessels and stimulates **aldosterone** release (adrenal cortex)
4	Aldosterone acts at the distal nephron/collecting duct → ↑ ENaC and Na^+/K^+-ATPase activity
5	Net response: **↑ Na^+ reabsorption; ↑ K^+ secretion** → restores ECF volume and BP

ANP (released with atrial stretch) generally promotes **natriuresis** (urinary Na^+ loss) — opposite effect to RAAS.

7 Potassium balance

K^+ is the major **intracellular** cation — essential for resting membrane potential, nerve and muscle function. Most filtered K^+ is reabsorbed earlier in the nephron; **variable secretion** occurs in the late distal tubule and collecting duct. High plasma K^+ → ↑ aldosterone → ↑ K^+ secretion.

8 Summary — ion handling at a glance

Ion	Main renal regulation	Physiological importance

Na^+	Reabsorption; RAAS / ANP	ECF volume and pressure
K^+	Distal secretion; aldosterone	Membrane excitability
Ca^{2+}	Reabsorption; PTH	Bone, contraction, signaling
HCO_3^- / H^+	Reabsorb HCO_3^-; secrete H^+	Acid–base balance

Phosphate and Mg^{2+} are also renally regulated; this lecture emphasizes Na^+, K^+, Ca^{2+} and acid–base ions.

P Must-remember pearls

- Water balance in steady state: **intake (≈2,500 mL/day: drinks + food + metabolic water) = output (≈2,500 mL/day: urine + skin + lungs + feces)**.
- The **thick ascending limb** reabsorbs $Na^+/K^+/2Cl^-$ via NKCC2 but is **impermeable to water** — this is what builds the medullary concentration gradient.
- **ADH** acts on the **V_2 receptor** → inserts **aquaporin-2** channels in the collecting duct → controls water reabsorption only, not sodium.
- **RAAS**: ↓renal perfusion → renin → angiotensin II → aldosterone → **↑Na^+ reabsorption, ↑K^+ secretion** at the distal nephron/collecting duct.
- Total-body Na^+ sets **ECF volume**; plasma $[Na^+]$ mainly reflects **water** balance relative to solute — these are two different things.

Q Test yourself

1. Which nephron segment is highly permeable to water but reabsorbs it only passively, with no active NaCl transport driving it?

A. Thick ascending limb
B. Loop of Henle descending limb
C. Distal convoluted tubule
D. Proximal convoluted tubule
E. Collecting duct

2. A patient becomes severely dehydrated. What happens to ADH secretion and the resulting urine?

A. ADH falls; large dilute urine results
B. ADH rises; small, concentrated urine results
C. ADH has no effect on urine concentration
D. ADH rises but urine becomes more dilute
E. ADH is irrelevant since aldosterone alone controls water

3. Which hormone, when released in response to low renal perfusion, ultimately increases Na^+ reabsorption and K^+ secretion at the distal nephron?

A. ADH
B. Atrial natriuretic peptide (ANP)
C. Aldosterone
D. Parathyroid hormone
E. Antidiuretic enzyme

4. A patient's atria are stretched by volume overload, triggering ANP release. What is the expected renal effect?

A. Increased Na^+ reabsorption (antinatriuresis)
B. Increased Na^+ excretion (natriuresis)
C. No renal effect — ANP acts only on the heart
D. Increased aldosterone release
E. Decreased GFR to zero

5. Why does the distal convoluted tubule's calcium reabsorption increase under PTH stimulation?

A. PTH increases DCT Ca^{2+} reabsorption, as described for the nephron's distal segment
B. PTH acts only on the proximal tubule
C. PTH decreases Ca^{2+} reabsorption everywhere
D. PTH has no role in renal calcium handling
E. PTH works only in the collecting duct

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — B — the descending limb is highly permeable to water and reabsorbs it passively; the thick ascending limb, by contrast, is impermeable to water.

2 — B — dehydration raises plasma osmolality, triggering ADH release, more aquaporin-2 insertion, and concentrated, low-volume urine.

3 — C — aldosterone, released via the renin-angiotensin pathway, increases ENaC and Na^+/K^+-ATPase activity at the distal nephron/collecting duct.

4 — B — ANP release with atrial stretch promotes natriuresis (urinary Na^+ loss), opposing RAAS.

5 — A — the lecture states the DCT reabsorbs Ca^{2+}, increased by PTH.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam

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Study alongside the lecture, not instead of it.